Project Report

on

AI Powered Job Recommendation System using ML

By

Umesh Mahajan – FYMCA61

Indira College of Engineering and Management, Pune

MCA-I

Introduction

The job search and recruitment process has largely shifted to online platforms, where thousands of job opportunities are available. However, finding the most suitable job based on a candidate's skills and qualifications remains challenging. Candidates often apply for multiple jobs without knowing if they match the requirements, while companies receive many irrelevant applications. This makes the recruitment process time-consuming and inefficient. An AI Powered Job Recommendation System uses Machine Learning to analyze candidate profiles and recommend suitable job opportunities based on skills, qualifications, and resume data. This helps improve the efficiency and accuracy of the recruitment process.

Existing System

In the current recruitment environment, many online job portals allow candidates to search and apply for jobs through web platforms. These systems mainly display job listings based on basic filters such as job title, location, and experience level. However, most of these systems do not analyze candidate resumes or skills intelligently to recommend suitable jobs. Candidates usually apply for multiple jobs manually without knowing whether their skills match the job requirements.

Limitations of Existing System:

- Job recommendations are not based on resume analysis or machine learning techniques.
- Candidates must manually search and apply for jobs.
- Recruiters receive many irrelevant job applications.
- Resume screening is mostly done manually.
- No automatic skill matching between candidate profiles and job descriptions.
- Limited or no integrated online interview verification system.
- Recruitment process becomes time-consuming and inefficient.

Need for the System

The need for this system arises due to the increasing demand for intelligent recruitment systems that can automate the job matching process.

An AI-based job recommendation system can significantly reduce the time required to search for relevant job opportunities. It analyses candidate skills and compares them with job requirements using machine learning algorithms.

- ✓ This system benefits both job seekers and recruiters by:
- ✓ Recommending jobs based on skills and qualifications
- ✓ Reducing irrelevant job applications
- ✓ Automating candidate screening
- ✓ Improving recruitment efficiency
- ✓ Providing an online interview verification system

Scope of Work

The scope of this project is to design and develop an intelligent web-based job portal that uses machine learning techniques to recommend suitable job opportunities to candidates based on their skills, resume data, and job requirements. The system aims to automate the recruitment process by providing intelligent job matching, resume analysis, and online interview verification…

Scope of the Work

- Develop a web-based job portal for job seekers and recruiters.
- Implement user registration and login system for secure access.
- Allow candidates to upload resumes and create professional profiles.
- Use machine learning techniques to analyze candidate skills and job descriptions.
- Recommend suitable jobs to candidates based on resume and skill matching.
- Provide job application functionality for candidates.
- Display match percentage between candidate skills and job requirements.
- Implement an online interview verification system using webcam.
- Generate interview scores based on candidate performance.
- Provide an HR/Admin dashboard to manage job postings and candidate applications.

- Allow recruiters to shortlist or reject candidates based on interview results.
- Maintain a database of users, job postings, applications, and interview resul

Operating Environment :

The proposed system is developed and executed in a web-based environment where users can access the application through a web browser.

Hardware Requirements :

- ✓ Processor: Intel Core i5 or higher (Minimum Intel i3 supported)
- ✓ RAM: Minimum 8 GB RAM
- ✓ Storage: Minimum 256 GB SSD / HDD
- ✓ Camera: Webcam for online interview verification
- ✓ Display: Standard monitor or laptop display
- ✓ Input Devices: Keyboard and Mouse
- ✓ Internet Connection: Required for accessing the web-based system

Software Environment / Software Requirements

Operating System: Windows 10

Programming Language: Python

Web Framework: Django

Frontend Technologies:

HTML5 – Page structure

CSS3 – Styling and layout design

Bootstrap – Responsive user interface design

JavaScript – Dynamic behavior and camera interaction

Database Management System: SQLite (Default Django Database)

Machine Learning Libraries:

NumPy – Numerical operations for machine learning processing

Pandas – Data analysis and dataset handling

Scikit-learn – Machine learning algorithms for job recommendation

Natural Language Processing (NLP) Techniques:

TF-IDF Vectorizer – Text feature extraction from resumes and job descriptions

Cosine Similarity – Similarity calculation for job matching and recommendation

Computer Vision Library: OpenCV – Webcam access and face detection for interview verification

Backend Processing: Django Views and Django ORM for handling server-side logic and database operations

API & Backend Handling:

Django Views and Django ORM are used to manage backend logic, handle user requests, and interact with the database for storing and retrieving application data.

Development Environment: Visual Studio Code (VS Code)

Python Environment: Virtual Environment (venv)

Web Browser: Google Chrome / Microsoft Edge

Software Detail Description of Technology Used

Python

Python is used as the primary programming language for developing the backend logic of the system. It is widely used in machine learning applications due to its simplicity and powerful libraries.

Django Framework

Django is used as the main backend framework for building the web application. It provides authentication, database connectivity, URL routing, and admin dashboard functionality.

HTML5

HTML5 is used to create the structure of the web pages such as login page, job listing page, dashboard, and interview interface.

CSS3

CSS3 is used to design and style the user interface of the system including layouts, colors, fonts, and responsive elements.

Bootstrap

Bootstrap framework is used to create responsive and modern UI components such as navigation bars, forms, buttons, cards, and dashboards.

JavaScript

JavaScript is used to implement dynamic features such as camera activation for interviews, form validations, and interactive web components.

SQLite Database

SQLite is used as the database management system for storing user data, job postings, applications, and interview results.

NumPy

NumPy is used for numerical computations and data processing required in machine learning operations.

Pandas

Pandas is used for handling structured data and processing datasets related to job descriptions and candidate profiles.

Scikit-learn

Scikit-learn is used to implement machine learning algorithms for job recommendation and similarity analysis.

TF-IDF Vectorizer

TF-IDF (Term Frequency – Inverse Document Frequency) is used to convert resume text and job descriptions into numerical feature vectors for machine learning analysis.

Cosine Similarity

Cosine Similarity is used to measure the similarity between candidate resume data and job descriptions to generate job match percentages

OpenCV

OpenCV library is used to access the webcam and perform face detection for online interview verification.

Virtual Environment (venv)

Virtual Environment is used to manage Python dependencies and isolate project libraries.

PROPOSED SYSTEM

Proposed System

The proposed system, AI Powered Job Recommendation System using Machine Learning, recommends suitable jobs to candidates based on their skills, resume data, and job requirements using machine learning and NLP techniques. Candidates can register, upload resumes, explore jobs, apply for positions, and attend online interviews through the platform with map-based company location viewing. In the proposed system, we have the following modules.

1. **User Management**

   Manages candidate registration and user profile information.

- User registration and login authentication
- User profile creation
- Resume upload and storage
- User dashboard management
- Password authentication and session handling

2. **Job Management**

   This module handles job postings and job information displayed on the platform

- Job creation and management
- Job listing and job details display
- Company information and job description management
- Job search and filtering by location or skills
- Job salary and experience display

3. **AI Job Recommendation**

   This module uses machine learning algorithms to recommend jobs to candidates.

- Resume text extraction from PDF
- Skill identification from resume
- Job description analysis
- Semantic similarity calculation using ML models
- Match percentage calculation
- Recommended job display for users

4. **Job Application Module**

   This module allows candidates to apply for jobs and track application status.

- Job application submission
- Resume attachment with application
- Skill match and missing skill analysis
- Application status tracking
- Candidate job history management

5. **Online Interview Verification Module**

   This module enables candidates to attend online interviews with verification.

- Camera access using webcam
- Face detection using OpenCV
- Online interview interface
- Interview completion process
- AI-based interview score generation

6. **Map and Location Module**

This module allows users to view company locations through map integration.

- Display company location on map
- Map marker for job location
- Latitude and longitude integration
- Company location view through map interface

7. **HR Dashboard Module**

This module allows HR managers to monitor and evaluate job applications.

- View candidate applications
- Resume download and review
- Candidate shortlist and rejection options
- Interview score evaluation
- Candidate selection monitoring

8. **Admin Dashboard Module**

This module allows administrators to control and manage the overall system.

- Job posting management
- User account management
- Application monitoring
- Recruitment analytics and statistics
- System data management

9. **Database Management Module**

This module manages all the system data stored in the database.

- User data storage
- Job information database
- Application records management

- Interview result storage
- System data updates and maintenance

Objectives of the System

The main objective of the proposed system is to develop an intelligent job recommendation platform that helps candidates find suitable job opportunities based on their skills, qualifications, and resume data.

To recommend relevant jobs to candidates based on their skills and resume data.

To automate the job matching process using Machine Learning techniques.

To allow candidates to upload resumes and apply for suitable job positions.

To help recruiters identify the most suitable candidates using match percentage analysis.

To provide an online interview verification system using webcam and face detection.

To display company job locations using map integration.

To provide Admin and HR dashboards for managing jobs, applications, and recruitment activities.

User Requirements

The system should provide a simple and efficient platform where candidates can easily search and apply for jobs while recruiters can manage applications and recruitment activities.

i. User Registration & Login – Allows users to create an account and securely access the system.
ii. Resume Upload – Enables users to upload resumes for skill analysis and job matching.
iii. Job Recommendation – Suggests suitable jobs to candidates based on their skills and resume data.
iv. Job Exploration – Allows users to browse available jobs with details such as company, salary, and location.
v. Map Location Viewing – Displays the company location on a map for better navigation.
vi. Job Application – Allows candidates to apply for jobs directly through the platform.
vii. Online Interview Verification – Enables candidates to attend interviews with webcam-based face detection.
i. User Dashboard – Allows users to track applied jobs and interview status.
ii. HR Management – Enables HR to review applications, analyze match scores, and approve or reject candidates.